

# C2\_00 Senior Analyst Foundations

Astra Parity RC4 Learner Book

**Eight senior-analyst habits taught from first principles: decision ownership, scoping, clarification, metric contracts, grain/lineage, judgment, validation, and handoff.**

**Route:** Learn -> practise -> mini-lab -> Quick Revision -> non-gating Module Test -> protected review -> Gate A.  
Gate B/C remain fresh retests only when assigned.

## How to use this module

Do not read this like a reference manual. For every lesson, follow the same evidence loop: understand the decision problem, reproduce the guided demonstration, save a fresh independent attempt, validate it, explain it without reading, and only then open the protected review.

| Lesson | Professional habit                                      |
|--------|---------------------------------------------------------|
| SAF1   | Junior analyst vs senior analyst responsibility         |
| SAF2   | Turning vague stakeholder requests into scoped analysis |
| SAF3   | Requirements gathering and clarification meetings       |
| SAF4   | KPI ownership, metric contracts and definitions         |
| SAF5   | Grain, lineage and source-of-truth decisions            |
| SAF6   | Assumptions, risks, limitations and analytical judgment |
| SAF7   | Professional validation and reconciliation              |
| SAF8   | Documentation another analyst can inherit               |

**Important: The protected Gate is competency based. A critical definition, grain, validation, or evidence-strength failure can block a pass even when the rest of the work looks polished.**

# SAF1 - Junior analyst vs senior analyst responsibility

## 1. Why it matters

A report can be technically correct and still damage a decision. Senior-level analysis means noticing what has not been defined, what could make the number wrong, and what another person needs in order to trust and reuse the work.

## 2. What you are learning

Separate task execution from decision ownership. Build a small ownership checklist that covers the decision, metric definition, grain/source, validation, risk communication, and handoff.

## 3. Prerequisite / what you already know

Course 1 basics: you can clean a small dataset, calculate a metric, and explain a chart. You do not need senior job experience.

## 4. Picture it in your head

**Decision guardrail: Picture a dashboard request arriving at a safety gate. A junior response asks, "Can I build it?" A senior response checks six locks before opening the gate: decision -> definition -> grain/source -> validation -> risk -> handoff. The software comes after the locks.**

## 5. New words before we use them

Decision owner: The person accountable for choosing the business rule or action. Acceptance criterion: A concrete check that tells you whether the analysis is good enough to use. Traceability: The ability to connect a conclusion back to definitions, source data, calculations, and decisions.

## 6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route.

## 7. Follow with me once

- Stakeholder says: 'Build the weekly sales dashboard by Friday.' A task-only response begins choosing charts.
- First ask what decision Friday's meeting must make. Assume the answer is: decide whether to change the weekend promotion.
- Write the minimum ownership contract: decision = promotion decision; metric = net revenue and margin; period = last 8 comparable weeks;
- grain = order line; exclusions = cancelled orders; sources = orders + returns + finance control.
- Add one acceptance criterion: net revenue must reconcile to the finance weekly control within the agreed tolerance before recommendation.

## 8. What you should see / be able to say

A short decision-protection checklist. It names the decision, definition, grain/source, validation check, and unresolved item before chart design begins.

## 9. Why it worked

The work is now judged by whether it supports a decision safely, not by whether a dashboard exists. Professional upgrade: In professional teams, senior analysts often spend more time reducing ambiguity, defining controls, documenting assumptions, and negotiating acceptance criteria than junior analysts expect.

## 10. Common beginner mistakes

1. Starting with chart colours or formulas. | 2. Treating the stakeholder's wording as a complete specification. | 3. Calling yourself 'senior' because you used an advanced tool. | 4. Hiding an unresolved definition so the work looks finished.

## 11. If you get stuck - recovery ladder

Write one sentence: 'What decision could be wrong if my analysis is wrong?' Then list the definition, source, and validation most likely to affect that decision. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

## 12. Now you do a small version alone

Fresh scenario: Operations asks, 'Why are refunds high? Send me something today.' Create a six-line ownership checklist before doing any calculation.

## 13. Validate your result

A different analyst should be able to answer: What decision is being supported? Which metric definition is still open? What evidence would block delivery?

## 14. Teach it back in your own words

Explain in 60-90 seconds why senior responsibility is not the same thing as using more advanced software.

## 15. Protected review + changed fresh retry

Save your genuine attempt. Only then open the protected SAF1 review. Close it and repeat on a changed scenario without copying wording.

## 16. Evidence / mastery + exact next route

Ownership checklist + one validation criterion + 60-90 second explain-back. Master only when you can reproduce the pattern on a new request. After saving evidence, continue to SAF2.

# SAF2 - Turning vague stakeholder requests into scoped analysis

## 1. Why it matters

Most real requests arrive incomplete: 'sales are down', 'customers are leaving', 'make a dashboard'. If you calculate before scoping, you can answer the wrong question perfectly.

## 2. What you are learning

Convert a vague request into an analysis brief: decision, audience, deadline, period, metric, grain, segments, exclusions, sources, and acceptance test.

## 3. Prerequisite / what you already know

SAF1 ownership mindset. Course 1 metric and filtering basics.

## 4. Picture it in your head

**Scoping funnel: Start with a wide, vague sentence at the top. Narrow it through decision -> audience -> KPI -> population/time -> grain/source -> deliverable -> acceptance. If any stage is blank, the request is still too wide to analyze safely.**

## 5. New words before we use them

Scope: The agreed boundary of what the analysis will and will not answer. Comparison period: The time period used as the reference for saying something increased or decreased. Exclusion: A record or population deliberately left out for a stated reason.

## 6. Watch / visual source

Scoped analysis brief + unresolved-question list + acceptance test + explain-back. RECOMMENDED VIDEO - requirements interview 00:00-05:42: problem

definition, gathering, refinement. Use for questioning technique; the book controls the course

<https://www.youtube.com/watch?v=iIAbSOvf96E>

After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report.

Save course-task evidence, not a screenshot of the source.

## 7. Follow with me once

- Do not calculate yet. Ask: 'Which decision will this analysis support?' Assume: decide whether to reduce the September promotion.

- Define the audience and deadline: Sales Director, Monday 10:00.

- Define the metric: net revenue after returns, not gross order value. Comparison: September vs August and vs same month last year.

- Declare grain and segments: order-line grain, region and product category. Exclude cancelled orders; keep returns as negative adjustments.

- Choose sources and acceptance: order lines + return events; reconcile monthly net revenue to Finance before recommendation.

## 8. What you should see / be able to say

A one-page analysis brief where a colleague could tell exactly what is being answered and what remains out of scope.

## 9. Why it worked

Each ambiguous word was converted into a decision or measurable contract before tools were used. Professional upgrade: A strong scope note also records what is explicitly out of scope and what would trigger a scope change. This protects both quality and delivery time.

## **10. Common beginner mistakes**

1. Using 'sales' without defining gross vs net. | 2. Comparing unequal time windows. | 3. Adding every possible segment instead of the ones relevant to the decision. | 4. Letting the deadline force silent assumptions.

## **11. If you get stuck - recovery ladder**

Circle every vague noun or adjective in the request: sales, active, high, recent, best, churned. Replace each with a definition or an open question. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

## **12. Now you do a small version alone**

Fresh scenario: Product says, 'Customers are churning after onboarding. Find the problem.' Build the scoped brief; do not calculate churn.

## **13. Validate your result**

Someone who did not attend the request meeting must be able to restate the decision, population, time window, metric, and exclusions from your brief.

## **14. Teach it back in your own words**

Explain why 'clarify before calculate' saves time even when the stakeholder says the request is urgent.

## **15. Protected review + changed fresh retry**

Save the independent brief first. Review only after attempt, then rebuild using a changed metric or time window.

## **16. Evidence / mastery + exact next route**

Scoped analysis brief + unresolved-question list + acceptance test + explain-back. RECOMMENDED VIDEO - requirements interview 00:00-05:42:

problem definition, gathering, refinement. Use for questioning technique; the book controls the course

<https://www.youtube.com/watch?v=iIAbSOvf96E>

After saving evidence, continue to SAF3.

# SAF3 - Requirements gathering and clarification meetings

## 1. Why it matters

A meeting is useful only if it changes uncertainty into recorded decisions. Beginners often ask many questions but fail to record who approved what, by when, and what happens if no answer arrives.

## 2. What you are learning

Plan a short clarification meeting, ask only analysis-changing questions, capture owners and decisions, and leave unresolved items visible.

## 3. Prerequisite / what you already know

SAF2 scoped analysis brief.

## 4. Picture it in your head

**Clarification tree: Imagine a meeting as a branching tree. Every answer creates the next useful question. Follow four branches - decision, definition, data, acceptance - and keep an "open / owner / due" log so uncertainty is visible rather than forgotten.**

## 5. New words before we use them

Elicitation: Drawing out information or requirements from stakeholders. Decision log: A dated record of questions, answers, owners, and unresolved items.

Escalation: Routing an unresolved decision to the person with authority to decide it.

## 6. Watch / visual source

Agenda + decision log + owner map + read-back summary + explain-back. RECOMMENDED VIDEO - requirements interview 00:50-05:42: problem definition through refinement.

<https://www.youtube.com/watch?v=iIAbSOvf96E>

After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

## 7. Follow with me once

- Prepare three questions only: Which revenue definition is approved? Are returns recognized on return date or original order date? Which regions are in scope?
- For every question, write why the answer can change the analysis. This prevents low-value questioning.
- During the meeting, record answer + owner + date. If the owner cannot decide, record the escalation owner.
- End by reading back the decisions and open items. Ask: 'If these are correct, may I proceed with this scope?'
- Send the decision log with the analysis brief. Do not rely on memory or chat history as the only record.

## 8. What you should see / be able to say

A meeting agenda and decision log with questions tied to analytical impact, named owners, status, and deadline.

## 9. Why it worked

The meeting produced an auditable contract instead of a collection of informal comments. Professional upgrade: In high-pressure work, the analyst may propose a temporary assumption, but it must be labelled, approved by the right owner, and paired with a trigger for recalculation.

## 10. Common beginner mistakes

1. Asking questions whose answers cannot change the work. | 2. Recording an answer without the owner who approved it. | 3. Leaving 'TBD' items with no deadline. | 4. Treating silence as approval.

### **11. If you get stuck - recovery ladder**

For each open item write: decision needed, impact if unresolved, owner, deadline, temporary assumption allowed? yes/no. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

### **12. Now you do a small version alone**

Fresh scenario: Marketing and Finance disagree on whether refunded orders count in campaign revenue. Prepare the agenda and decision log.

### **13. Validate your result**

Every open question has an owner and a next action. Every resolved question is reflected in the scope or metric contract.

### **14. Teach it back in your own words**

Explain the difference between 'asking more questions' and 'reducing decision uncertainty'.

### **15. Protected review + changed fresh retry**

Attempt first. Then inspect the protected review and retry with a different stakeholder conflict.

### **16. Evidence / mastery + exact next route**

Agenda + decision log + owner map + read-back summary + explain-back. RECOMMENDED VIDEO - requirements interview 00:50-05:42: problem definition through refinement. <https://www.youtube.com/watch?v=iIAbSOvf96E> After saving evidence, continue to SAF4.

# SAF4 - KPI ownership, metric contracts and definitions

## 1. Why it matters

Two teams can show different numbers while both calculations are arithmetically correct. The disagreement is often a definition problem:

different populations, dates, exclusions, or sources.

## 2. What you are learning

Build a versioned metric contract for a KPI and distinguish label, business meaning, formula, eligible population, exclusions, grain, time basis, owner, source, target, and tolerance.

## 3. Prerequisite / what you already know

SAF2 scope and SAF3 decision ownership.

## 4. Picture it in your head

**Metric nutrition label: A KPI name is only the product name on the front. The contract is the label on the back: formula, numerator, denominator, population, exclusions, time basis, grain, source, owner, version. Two teams using different labels do not have the same metric.**

## 5. New words before we use them

KPI: A metric tied to an important business objective or decision. Denominator: The population used underneath a rate or percentage. Metric contract: A

documented agreement defining exactly how a metric is interpreted and calculated.

## 6. Watch / visual source

Versioned metric contract + one reconciliation rule + explain-back. RECOMMENDED REFERENCE - IIBA KPI guidance Use for KPI framing; course contract fields are authoritative for assessment.

<https://www.iiba.org/professional-development/knowledge-centre/articles/kpis-for-business-analysts/>

After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report.

Save course-task evidence, not a screenshot of the source.

## 7. Follow with me once

- Net Revenue meaning: revenue recognized from completed order lines after approved returns and discounts.
- Formula:  $\text{sum}(\text{line\_amount} - \text{approved\_return\_amount} - \text{discount\_amount})$ . Grain: order line. Time basis: order completion month.
- Return Rate meaning: returned completed order lines divided by eligible completed order lines. State whether partial returns count by line or
- order.
- Name exclusions: cancelled orders, test orders, fraudulent orders if approved. Name owner: Finance for revenue recognition; Operations may
- own operational return reason taxonomy.
- Add version date and tolerance: e.g. reconcile monthly Net Revenue to Finance control within 0.5% or investigate before publishing.

## 8. What you should see / be able to say

A metric contract where another analyst could implement the same KPI without guessing hidden rules.

## 9. Why it worked

The metric is treated as a business agreement first and a formula second. Professional upgrade: Organizations often separate metric owner, data owner, and report owner. Do not assume the dashboard developer has authority to define the business metric.

## 10. Common beginner mistakes

1. Writing only the formula. | 2. Ignoring the denominator of a rate. | 3. Using dashboard label as definition. | 4. Changing a metric silently when the source changes.

## 11. If you get stuck - recovery ladder

If two numbers disagree, compare contract fields one by one: meaning, population, exclusions, date basis, grain, source, version. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

## 12. Now you do a small version alone

Fresh scenario: Define 'On-time Delivery Rate' for an executive dashboard. Include numerator, denominator, exclusions, time basis, grain, owner, source, target, and version.

## 13. Validate your result

Ask a second analyst to independently restate the numerator and denominator. If they cannot, the contract is not yet clear.

## 14. Teach it back in your own words

Explain why a KPI can be wrong even when the formula contains no error.

## 15. Protected review + changed fresh retry

Attempt first, then compare against protected signals and retry with a different rate metric.

## 16. Evidence / mastery + exact next route

Versioned metric contract + one reconciliation rule + explain-back. RECOMMENDED REFERENCE - IIBA KPI guidance Use for KPI framing; course contract fields are authoritative for assessment.  
<https://www.iiba.org/professional-development/knowledge-centre/articles/kpis-for-business-analysts/>  
After saving evidence, continue to SAF5.

# SAF5 - Grain, lineage and source-of-truth decisions

## 1. Why it matters

Many believable BI errors come from combining tables that describe different things: one row per order, one row per order line, one row per return event, one row per monthly target. If the grain is not declared, joins can multiply values silently.

## 2. What you are learning

Declare grain and keys, map lineage from source to output, and choose the source-of-truth owner for each business fact.

## 3. Prerequisite / what you already know

Course 1 joins/model basics plus SAF4 metric contract.

## 4. Picture it in your head

**One-row map: Put a sticky note above every table saying exactly what one row means. Draw arrows only after that. Then mark one source as authoritative for each business fact. If two sources disagree, the map must show who decides - not silently pick the easier number.**

## 5. New words before we use them

Grain: What exactly one row represents. Lineage: The trace of where data came from and how it changed before reaching the output. Source of truth: The agreed authoritative system or owner for a particular business fact.

## 6. Watch / visual source

Source/grain register + lineage map + join-risk note + explain-back. RECOMMENDED CURRENT REFERENCE - Power BI data lineage Use the lineage-view idea as a visual analogy; no Power BI service access is required for this lesson.  
<https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-data-lineage>  
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

## 7. Follow with me once

- orders: one row per order. Key = order\_id.
- order\_lines: one row per order + line number. Key = order\_id + line\_id. Revenue is naturally calculated here.
- returns: one row per return event. One line can have multiple return events, so joining directly without aggregation can duplicate line revenue.
- targets: one row per region-month. It is a higher-grain target and should not be joined blindly to each order line for aggregation.
- Write lineage: source tables -> cleaning rules -> return aggregation -> metric calculation -> executive report. State which system owns each
- fact.

## 8. What you should see / be able to say

A source/grain register plus a small lineage chain. Every table has a declared row meaning and key.

## 9. Why it worked

You prevented fan-out by deciding what each row means before joining or aggregating. Professional upgrade: Source-of-truth is fact-specific. Finance can own recognized revenue while CRM owns customer segment and Operations owns fulfillment status.

## 10. Common beginner mistakes

1. Calling a file 'customer level' when it actually has multiple rows per customer. | 2. Choosing source of truth because it is easiest to access. | 3. Assuming a primary key without checking duplicates. | 4. Using DISTINCT to hide duplicated joins.

## 11. If you get stuck - recovery ladder

Before any join, write two sentences: 'One row in left table means...' and 'One row in right table means...'. Predict the output grain before running the join. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

## 12. Now you do a small version alone

Fresh scenario: You have ticket, ticket\_event, agent, and monthly\_team\_target tables. Declare grain/key and draw the lineage for 'resolved tickets per team vs target'.

## 13. Validate your result

For every join, state expected row count direction and one duplicate/unmatched-key check.

## 14. Teach it back in your own words

Explain grain without using the word 'granularity'.

## 15. Protected review + changed fresh retry

Save your source/grain register before opening the protected review. Fresh retry must use different table grains.

## 16. Evidence / mastery + exact next route

Source/grain register + lineage map + join-risk note + explain-back. RECOMMENDED CURRENT REFERENCE - Power BI data lineage Use the

lineage-view idea as a visual analogy; no Power BI service access is required for this lesson.

<https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-data-lineage> After saving evidence, continue to SAF6.

# SAF6 - Assumptions, risks, limitations and analytical judgment

## 1. Why it matters

Decision-makers need to know not only what you found but how confident they should be. Mixing facts, assumptions, and recommendations makes a conclusion sound stronger than the evidence.

## 2. What you are learning

Label facts, assumptions, risks, limitations, and recommendations separately; calibrate recommendation strength to evidence.

## 3. Prerequisite / what you already know

SAF1-SAF5 decision, scope, metric, and source controls.

## 4. Picture it in your head

**Four-box judgment board: Separate what you observed, what you assumed, what could go wrong, and what the data cannot establish. Your recommendation is allowed to be only as strong as the weakest box that materially affects the decision.**

## 5. New words before we use them

Assumption: Something treated as true for the analysis but not fully proven. Risk: A possible future or hidden condition that could make the result wrong or harmful. Limitation: A known boundary of the data or method. Trigger: New evidence or a threshold that would change the recommendation.

## 6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route.

## 7. Follow with me once

- Fact: East net revenue is 12% below the approved target for the current month after reconciliation.
- Assumption: the promotion calendar supplied by Marketing is complete.
- Limitation: offline store orders arrive one day later and are not yet included.
- Risk: if late offline orders are unusually high, the 12% gap may shrink materially.
- Recommendation: investigate the promotion mix before expanding discounts. Trigger: recompute after offline close; reverse the
- recommendation if the gap falls below the agreed materiality threshold.

## 8. What you should see / be able to say

A decision note where evidence strength is obvious and the reader knows what could change the conclusion.

## 9. Why it worked

The recommendation is neither timid nor overconfident; it is tied to evidence and a change trigger. Professional upgrade: Senior judgment often means choosing the safest decision language: 'evidence supports', 'directional only', 'do not act until', or 'act now with monitoring' based on consequence and uncertainty.

## 10. Common beginner mistakes

1. Calling an assumption a fact. | 2. Using 'probably' without naming the uncertainty. | 3. Listing limitations but ignoring them in the recommendation. | 4. Giving an irreversible action from weak evidence.

### **11. If you get stuck - recovery ladder**

Highlight every sentence in your recommendation. Label it F/A/R/L/REC. Any sentence you cannot classify needs rewriting. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

### **12. Now you do a small version alone**

Fresh scenario: Customer satisfaction fell after a support-tool migration, but survey response rate also changed. Write fact/assumption/risk/limitation/recommendation + trigger.

### **13. Validate your result**

A reviewer should be able to identify exactly which new evidence could strengthen, weaken, or reverse the recommendation.

### **14. Teach it back in your own words**

Explain why a limitation does not automatically make an analysis useless.

### **15. Protected review + changed fresh retry**

Attempt first. Retry on a scenario where the data limitation is different.

### **16. Evidence / mastery + exact next route**

Calibrated decision note + trigger + explain-back. After saving evidence, continue to SAF7.

# SAF7 - Professional validation and reconciliation

## 1. Why it matters

A dashboard can look polished while containing a duplicated join, missing period, stale file, or wrong denominator. Validation is not a final glance; it is a planned second path that tries to prove the result wrong.

## 2. What you are learning

Design validation before delivery using row counts, coverage, duplicates, unmatched keys, control totals, independent recomputation, business invariants, tolerances, and recovery.

## 3. Prerequisite / what you already know

Course 1 data-cleaning checks plus SAF4 metric contract and SAF5 grain.

## 4. Picture it in your head

**Reconciliation triangle: Put source control, transformed data, and final output at three corners. Important counts/amounts must travel around the triangle without an unexplained jump. If one edge fails, stop and repair before presenting.**

## 5. New words before we use them

Reconciliation: Comparing results against an independent control or alternate calculation. Tolerance: The maximum acceptable difference before investigation is required. Invariant: A rule expected to remain true if the data and logic are correct.

## 6. Watch / visual source

Validation log + independent control + failure-recovery note + explain-back. RECOMMENDED CURRENT REFERENCE - Power BI data profiling Use the profiling section as visual reinforcement for detecting unexpected/null values; course validation plan is b <https://learn.microsoft.com/en-us/training/modules/clean-data-power-bi/>

After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

## 7. Follow with me once

- Input check: confirm expected files and date coverage are present before calculating.
- Key check: detect duplicate order-line keys and unmatched return events.
- Control total: compare completed gross revenue to the operational source control.
- Independent recompute: calculate Net Revenue a second way using a separate grouped summary or pivot, not the same formula copied twice.
- Business invariant: Net Revenue should not exceed eligible gross revenue. Record tolerance and recovery steps for every failed check.

## 8. What you should see / be able to say

A validation plan with expected values or control methods, observed result, tolerance, status, and recovery path.

## 9. Why it worked

Different failure types have different checks; one green total cannot prove every part of the pipeline. Professional upgrade: For important outputs, store validation evidence with the release so future analysts can see what was checked, what tolerance was accepted, and what failed.

## 10. Common beginner mistakes

1. Validating with the same formula that produced the result. | 2. Ignoring a small mismatch without an agreed tolerance. | 3. Fixing the data silently instead of recording the cause. | 4. Treating 'no error message' as validation.

## 11. If you get stuck - recovery ladder

Ask: What wrong result could still look plausible? Design one check specifically to catch that failure. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

## 12. Now you do a small version alone

Fresh scenario: Validate a support dashboard showing resolved ticket rate by team. Design at least six checks before seeing the final dashboard.

## 13. Validate your result

At least one check must be independent of the main calculation and at least one must test grain/key integrity.

## 14. Teach it back in your own words

Explain the difference between checking that code runs and checking that a business result is trustworthy.

## 15. Protected review + changed fresh retry

Attempt first. After review, design a changed validation plan for a different metric and source structure.

## 16. Evidence / mastery + exact next route

Validation log + independent control + failure-recovery note + explain-back. RECOMMENDED CURRENT REFERENCE - Power BI data profiling Use the profiling section as visual reinforcement for detecting unexpected/null values; course validation plan is b <https://learn.microsoft.com/en-us/training/modules/clean-data-power-bi/> After saving evidence, continue to SAF8.

# SAF8 - Documentation another analyst can inherit

## 1. Why it matters

If only you can refresh or explain the work, the team owns a fragile dependency. Senior-quality documentation lets another analyst reproduce the output and know when not to trust it.

## 2. What you are learning

Create an inheritable handoff pack covering purpose, definitions, sources/access, grain/keys, steps, validation, limitations, refresh/run instructions, owners, escalation, and change history.

## 3. Prerequisite / what you already know

SAF1-SAF7. You need one completed small analysis artifact to document.

## 4. Picture it in your head

**Bus-factor test: Imagine you disappear for two weeks and a new analyst receives only your handoff. Can they find the source, definitions, assumptions, refresh steps, validation evidence, known limitations, and next action without messaging you? If not, the analysis is not inheritable.**

## 5. New words before we use them

Runbook: Step-by-step instructions for operating or refreshing a repeatable process. Change log: A dated history of material changes and why they were made. Escalation owner: The person to contact when a rule, source, or process cannot be resolved by the analyst.

## 6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route.

## 7. Follow with me once

- Purpose: weekly executive check on net revenue, margin, and return rate; decision owner = Commercial Director.
- Definitions: link to metric-contract version and state effective date.
- Sources: exact source names, refresh timing, access owner, expected file/table grain.
- Run steps: numbered sequence from source refresh through transformation, validation, report refresh, and evidence save.
- Checks and limitations: list required reconciliations, tolerances, known delayed source, and what to do if a control fails.
- Ownership: report owner, metric owner, data owner, escalation path, and change history.

## 8. What you should see / be able to say

A new analyst can reproduce the normal run and recognize a failure without asking you what each step means.

## 9. Why it worked

The handoff documents decisions and failure handling, not just button clicks. Professional upgrade: Good documentation is concise but executable. Keep authoritative definitions in one controlled location and reference them instead of copying conflicting versions into many files.

## 10. Common beginner mistakes

1. Writing 'refresh data' without naming the source or expected timing. | 2. Documenting only the happy path. | 3. Leaving metric definitions inside your head. | 4. Putting passwords or secrets into documentation.

### **11. If you get stuck - recovery ladder**

Give the handoff to an imaginary analyst who has a clean folder and no chat history. Mark every sentence where they would need to ask you a question. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

### **12. Now you do a small version alone**

Fresh scenario: Document a monthly customer-retention report so a new analyst can run it next month. Include failure and escalation paths.

### **13. Validate your result**

Use the cold-handoff test: another person should identify inputs, outputs, definitions, steps, validations, limitations, and owners from the document alone.

### **14. Teach it back in your own words**

Explain why documentation is part of analytical quality rather than administrative work.

### **15. Protected review + changed fresh retry**

Attempt first, then compare against the protected review and rebuild one section from a changed failure scenario.

### **16. Evidence / mastery + exact next route**

Handoff pack + cold-handoff checklist + explain-back. After saving evidence, complete the module mastery checklist and follow the Gate A / completion route.